Chapter 1

Introduction

The Roommate Finder Website is a web-based application developed to help users find suitable roommates and shared accommodation easily. In many cities, students and working professionals move frequently for education and jobs, and they often need shared rooms to reduce living expenses. Finding a compatible roommate and a suitable room becomes difficult using traditional methods.

Currently, people depend on brokers, friends, notice boards, and social media platforms to search for shared accommodation. These methods are not organized and often provide incomplete or outdated information. Users also face problems such as fake listings, lack of proper contact details, and difficulty in comparing available rooms. Because of this, the process becomes time- consuming and inefficient.

There is a need for a centralized system that allows users to search for rooms and find roommates in a simple and structured way. A digital platform can reduce manual effort and provide verified information. This helps users save time and make better decisions.

The proposed system provides an online platform where users can search for rooms based on their preferences. Users can view room details such as rent, location, facilities, and roommate requirements. This helps users select rooms according to their needs.

The system also allows users to post room availability. Users can add details such as room type, rent amount, location, and contact information. This makes it easier for others to find available rooms quickly.

The application includes a search feature that helps users filter rooms based on requirements. This improves usability and makes the system more efficient. Users can also view detailed room information before contacting the owner.

An admin module is provided to manage the system. The admin can verify room posts, remove invalid entries, and monitor overall system activity. This improves transparency and reduces fake or duplicate listings.

The system is developed using modern web technologies to provide a responsive and user- friendly interface. The frontend handles user interaction, while the backend manages data processing and storage. The database stores user and room information securely.

The main objective of this system is to simplify the process of finding roommates and shared accommodation. The system reduces manual work, saves time, and improves communication between users. Overall, the Roommate Finder provides an efficient solution for shared living arrangements.

Chapter 2
Literature Survey

2.1 Study of Existing System

We studied the existing system related to finding roommates and shared accommodation. In many cities, people still depend on brokers, friends, and social media groups to find rooms and roommates.

We interacted with students and working professionals to understand the current room searching process. During the discussion, the following questions were asked.

Q1. How do you search for rooms or roommates?
Q2. Do you find complete room details easily?
Q3. Do you depend on brokers for room searching?
Q4. What problems occur in manual searching?
Q5. Is there any proper online platform available?
Q6. How do you contact room owners or Roommate?
Q7. Do you face difficulty in comparing multiple rooms?

After discussing with them, we learnt that :-

Ans 1: Most users search for rooms using brokers, social media posts, or personal contacts. These methods are manual and not properly organized.

Ans 2: No, users do not always get complete room details such as rent, facilities, and location. Information is often incomplete or unclear.

Ans 3: Yes, many users depend on brokers for finding rooms or roommates. This increases cost and sometimes provides limited options.

Ans 4: Manual searching creates problems such as fake listings, incomplete details, and time- consuming communication. It is also difficult to compare multiple rooms.

Ans 5: Some online platforms are available but many do not provide proper verification. They also lack structured search and filtering features.

Ans 6: Users usually contact room owners through phone numbers shared in posts. Communication is not centralized or properly managed.

Ans 7: Yes, users face difficulty in comparing rooms because details are not organized. This makes the decision-making process harder.

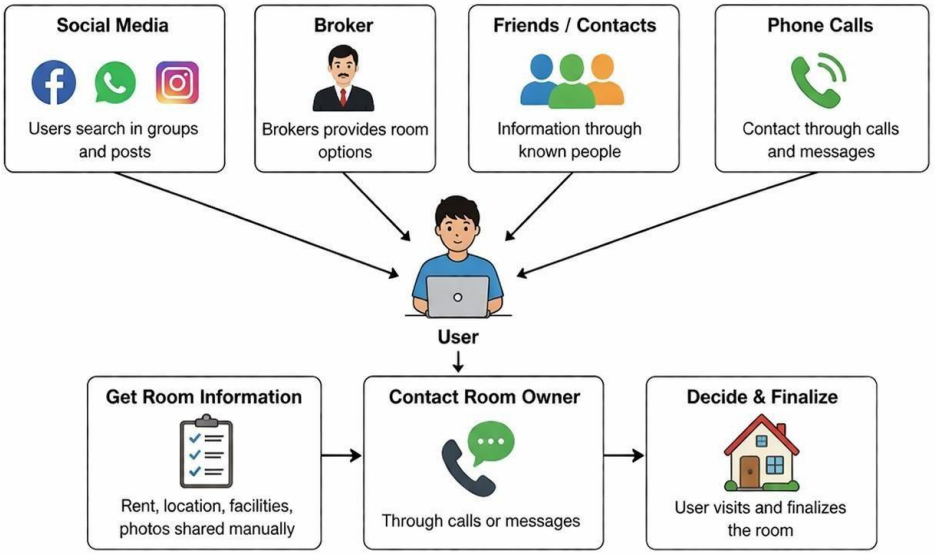

Fig 2.1 Existing Room Searching Method

The above figure shows the existing method used for finding rooms or roommates. In this system, users search for rooms using brokers, social media platforms, and personal contacts. Room details such as rent, location, and facilities are shared manually through phone calls or messages.

This method is simple but has several limitations. There are chances of fake information, incomplete details, and difficulty in comparing multiple rooms. It is also time- consuming and not properly organized. Users may miss suitable room opportunities due to lack of a centralized platform.

2.1 Limitations of Existing System

- The existing system used for finding roommates and shared accommodation is mostly manual and unorganized. Users depend on brokers, social media posts, and personal contacts to search for rooms. Although these methods are commonly used, they have several limitations that affect efficiency and reliability.

- One of the major problems is that users do not get complete and verified information. Room details such as rent, location, and facilities are often shared informally. This may lead to confusion and incorrect decisions.

- Another limitation is the absence of a centralized platform. Information is scattered across different sources such as social media groups and phone contacts. This makes it difficult for users to compare multiple room options.

- The system is also time-consuming. Users need to search in different places, contact multiple people, and manually collect information. This increases effort and delays the decision-making process.

- There are also chances of fake or outdated listings. Since there is no verification process, some room details may not be accurate. Users may waste time contacting unavailable or incorrect listings.

- Communication between users and room owners is not properly managed. Users depend on phone calls and messages, which makes tracking conversations difficult. There is also no proper record of communication.

- Another limitation is difficulty in filtering rooms based on requirements. Users cannot easily search rooms based on rent, location, or facilities. This makes the searching process less efficient.

Summary of Limitations

- Incomplete and unverified room information
- No centralized platform for room searching
- Time-consuming manual searching process
- Chances of fake or outdated listings
- Poor communication management
- Difficulty in filtering room details
- No admin verification or control

2.3 Problem Identification

The study of the existing system shows that there are several problems faced by users while searching for roommates and shared accommodation. These problems arise mainly due to manual searching methods and lack of a centralized platform.

Problems of User Related Issues

- Users do not have a proper platform to search rooms easily
- Users do not get complete room details in one place
- Difficulty in comparing multiple room options
- Users depend on brokers or personal contacts
- Time-consuming process for finding suitable roommates

Problems of Room Owner Related Issues

- Room owners cannot easily reach interested users
- Difficulty in posting room availability to a large audience
- No proper system to manage room details
- Hard to update or remove room information

Problems of Admin / System Related Issues

- Room owners cannot easily reach interested users
- Difficulty in posting room availability to a large audience
- No proper system to manage room details
- Hard to update or remove room information
- Communication with multiple users becomes difficult

Problems of Existing System (Manual System)

- No centralized platform for room searching
- Incomplete or incorrect room information
- Time-consuming manual communication
- Chances of fake or outdated listings
- Difficulty in filtering room details
- No proper admin control

Overall Problems

- Lack of proper digital platform for roommate finding
- Difficulty in searching and comparing rooms
- Unorganized room information
- Inefficient communication between users
- Time-consuming and unreliable system

Table No: 2.1 Problem Analysis

Category	**Problems**
User	Difficulty in searching rooms, incomplete details, time-consuming
Room Owner	Hard to post rooms, manage data, communication issues
Admin	No verification, difficult to manage listings
System	No database, fake listings, no centralized platform

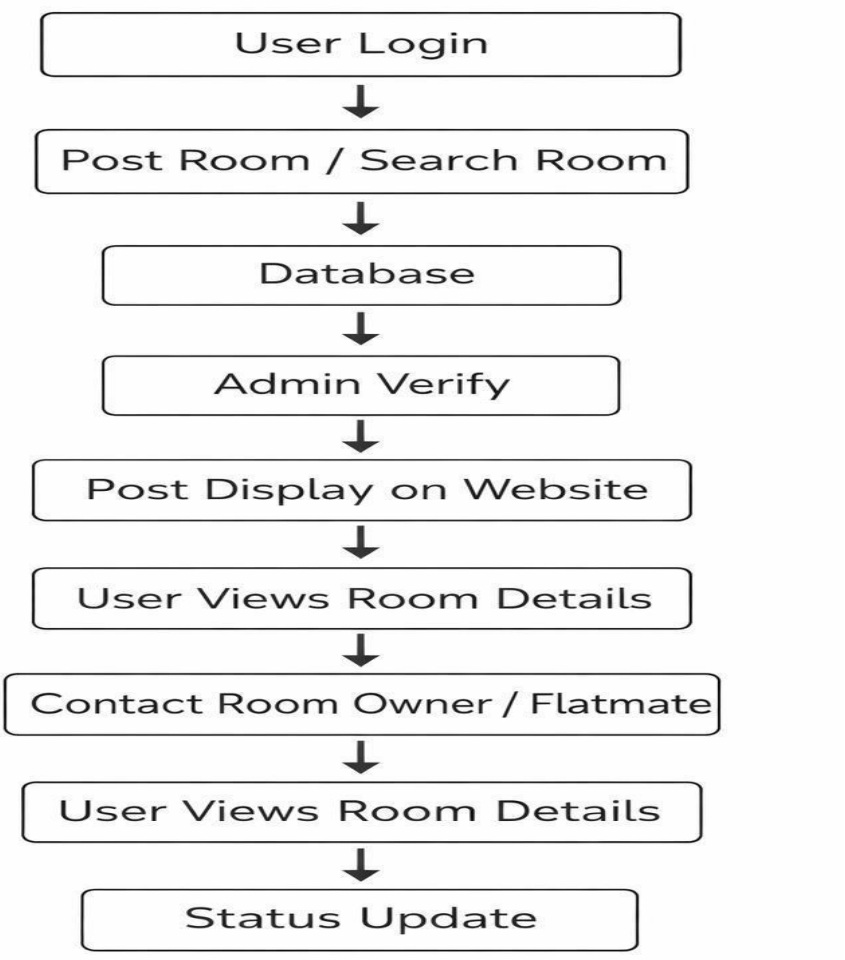

Fig 2.3 Problem Identification Flow

The above figure shows how the roommate finder system works Users first log into the system and then post room details or search for available rooms. The submitted data is stored in the database and verified by the admin before being displayed on the website.

2.2 Problem Definition

The existing system for finding roommates and shared accommodation is mostly manual and unorganized, which creates several challenges for users and room owners. Due to these limitations, there is a need to develop an efficient and reliable system that can manage room searching and posting in a better way.

The main objective of this project is to develop a web-based Roommate Finder that will help users find rooms and roommates easily. The proposed system aims to

The system will focus on:

a) Maintaining proper records related to room details and roommate information

b) Managing room availability and search in an organized manner

c) Providing easy access to room data for users

To achieve this, the proposed system will provide:

a) **Detailed categorization of room records:**
The system will store room data in a structured format including location, rent, facilities, and contact details. This will help in organizing data properly.

b) **Provision to maintain room details**:
The system will allow users to add, update, and view room details easily. This will reduce manual work and improve data accuracy.

c) **Search and filtering system:**
The system will provide search options to find rooms based on requirements such as rent, location, and facilities.

d) **Centralized database management:**
All records will be stored in a centralized database, making it easy to access data anytime and from anywhere.

e) **Admin verification:**
The admin will verify room posts before displaying them on the website. This helps reduce fake listings.

Benefits of Proposed System

- Reduces manual searching effort
- Improves accuracy of room information
- Saves time for users
- Provides centralized platform for room searching
- Improves communication between use

Chapter 3

Scope of the Project

3.1 Scope of the project

The Roommate Finder Website is designed to provide an easy and efficient way for users to find roommates and shared accommodation. The scope of this project includes developing a web-based platform where users can search for rooms, post room availability, and contact potential roommates.

The project focuses on providing a centralized platform for managing room details and user information. Users can register in the system, log in, and search for available rooms based on their requirements. They can also post room availability with details such as location, rent, facilities, and contact information. This helps users find suitable roommates quickly.

The system also includes an admin module to manage room postings and user data. The admin verifies room details before displaying them on the website. This improves transparency and reduces fake or duplicate listings.

Key Functional Areas of the System

The scope of this project is divided into several functional modules which help in efficient management of the system.

1) User Registration and Login

- The system allows users to register using basic details such as name, email, and password.
- Each user will have a unique account in the system.
- Users can securely log in to access system features.
- The system allows users to update and manage their profile details.

2) Search Room Module

- The system allows users to search available rooms based on their requirements.
- Users can view room details such as location, rent, and facilities.
- The search feature helps users compare different room options.

- It reduces manual searching effort and saves time.

3) Post Room Module

- Users can post room availability by entering room details.
- Details such as rent, location, and facilities are stored in the system.
- Posted rooms are saved in the database.
- This helps others find available rooms easily.

4) View Room Details

- The system displays complete room information to users.
- Users can check details before contacting the owner.
- This improves decision making.
- It helps users compare multiple rooms effectively.

5) Contact Owner / Roommate

- Users can contact room owners using provided contact details.
- This improves communication between users.
- It reduces dependency on brokers.
- Users can directly discuss room availability.

6) Admin Verification and Management

- Admin verifies room posts before displaying them.
- Admin can manage users and room data.
- Invalid or fake posts can be removed.
- This improves reliability of the system.

7) Data Storage and Security

- All data is stored in a centralized database.
- It ensures safe and organized data management.
- Secure login protects user information.
- Data can be accessed easily when required.

8.Benefits of the Proposed System

- Reduces manual searching effort
- Improves accuracy of room information
- Saves time for users
- Provides centralized platform
- Improves communication between users
- Reduces dependency on brokers

9.Future Scope of the Project

The system can be further enhanced with additional features such as:

- Mobile application for easy access
- Chat system for user communication
- Map integration for room location
- Advanced search and filtering options
- Cloud-based storage for better accessibility

Chapter 4

Methodology

4.1.1 Requirements Gathering and Analysis

For the development of the Roommate Finder Website, we have adopted the Waterfall Model as the software development approach. The Waterfall Model is a linear and sequential process where each phase is completed before moving to the next phase. This model is suitable for this project because the requirements are clearly defined and structured Requirements Gathering and Analysis

The Software Development Life Cycle (SDLC) followed in this project includes the following phases:

1. Requirements Gathering and Analysis
2. System Design
3. Testing
4. Implementation
5. Deployment

Phase 1: Requirements Gathering and Analysis

In this phase, we focused on understanding the needs of users who want to find rooms and users who want to post rooms. The main objective was to identify the problems in the existing system and define the requirements of the proposed system.

We collected information through:

- Discussion with students and working professionals
- Observation of existing room searching methods
- Analysis of online room finder platforms

This helped us to design a system that solves real-world problems effectively.

4.1.2 User Requirements

The system should fulfill the following requirements for users:

- Users should be able to register and login into the system
- Users should be able to search available rooms
- Users should be able to view room details
- Users should be able to contact room owner or flatmate
- Users should be able to post room details
- The system should be simple and user-friendly

4.1.3 Admin Requirements

- Admin should be able to verify room posts
- Admin should manage users and room listings
- Admin should remove invalid or fake posts
- Admin should monitor system activities
- Admin should manage database records

4.1.4 Room Posting Requirements

- The system should allow users to post room details
- It should store rent, location, and facilities
- Each room should have a unique record
- The system should store room images and description
- Posted rooms should be verified before display

4.1.5 Room Search Requirements

- The system should allow searching rooms by location
- Users should view available room details

• It should display rent and facilities information

• Users should compare different rooms

• It should provide easy navigation

4.1.6 Record Management Requirements

• The system should store all data in structured format

• It should allow adding and updating room records

• It should maintain user and room data

• It should provide easy access to stored data

• It should store data securely in database

4.1.7 System Requirements Software Requirements

- Operating System: Windows OS
- Platform: Visual Studio Code
- Languages: HTML, CSS, JavaScript, React
- Backend: Node.js
- Database: Mongo DB

- **Phase 1:Hardware Requirements**

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: 500 GB or more

- **Phase 2: System Design**

In this phase, the system architecture, database structure, and user interface were designed.

• Database design includes tables for users, babies, and vaccination records

- User interface is designed for easy navigation
- System modules are defined clearly

- **Phase 3: Implementation**

  In this phase, the system is developed using:

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js
- Database: Mongo DB

- **Phase 4: Testing**

  Testing is done to ensure the system works correctly.

- Unit Testing – testing individual modules
- Integration Testing – testing combined modules
- System Testing – testing the complete system Errors and bugs are identified

- **phase 5: Deployment**

- The system is deployed on a server
- Users can access the system through a web browser
- System becomes ready for real-time use

- **Phase 6: Maintenance**

- Regular updates are performed
- Bugs are fixed if found
- System performance is improved over time

Chapter 5

Details of Design, Working and Processes

Phase 2: System Design

In the System Design phase of the Roommate Finder Website, the main objective was to convert the gathered requirements into a structured and well-organized system. This phase plays a crucial role in defining how the system will function and how different components will interact with each other.

The system design includes:

- System Architecture
- Database Design
- Module Design
- User Interface Design

The design ensures that the system is efficient, user-friendly, and capable of handling all required operations smoothly.

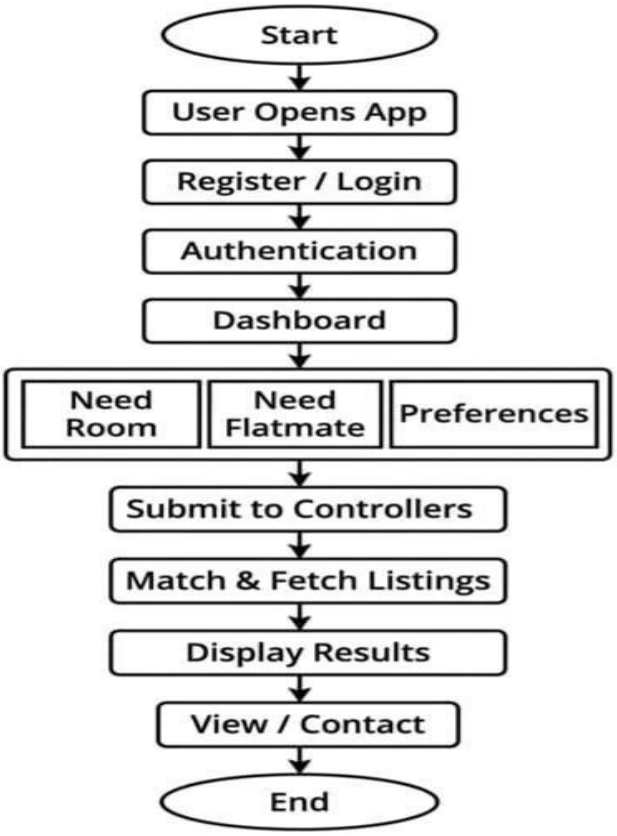

Fig 5.1: System Workflow Diagram

The above figure represents the overall working of the Roommate Finder Website, which involves three main users: User, Room Owner, and Admin. The system is designed to ensure smooth communication and efficient management of room searching and posting processes.

Initially, the process starts with the user. The user registers into the system, logs in, and searches for available rooms. Based on the requirements such as location, rent, and facilities, the user can view room details and select a suitable option.

Once the room is selected, the user can contact the room owner using the provided contact details. The room owner can provide additional information and confirm availability. Users can also post their own room details if they want to find a roommate or tenant.

After posting the room, the details are stored in the system database. The admin verifies the posted room information and approves it. Once approved, the room becomes visible to all users searching for rooms.

The admin plays a vital role in managing the entire system. The admin is responsible for managing users, verifying room posts, removing invalid listings, and monitoring system activities. The admin also manages the database and ensures smooth functioning of the system.

Step-wise Working of System

1. User registers and logs into the system
2. User searches for available rooms
3. System displays room list with details
4. User selects a room and views full information
5. User contacts room owner / Roommate
6. Room owner provides additional details
7. User can post room details if required
8. Room information is stored in database
9. Admin receives new room posting request
10. Admin verifies room details
11. Admin approves or rejects post

12. Approved rooms are displayed to users
13. Users view and compare available rooms
14. Admin manages entire system and database

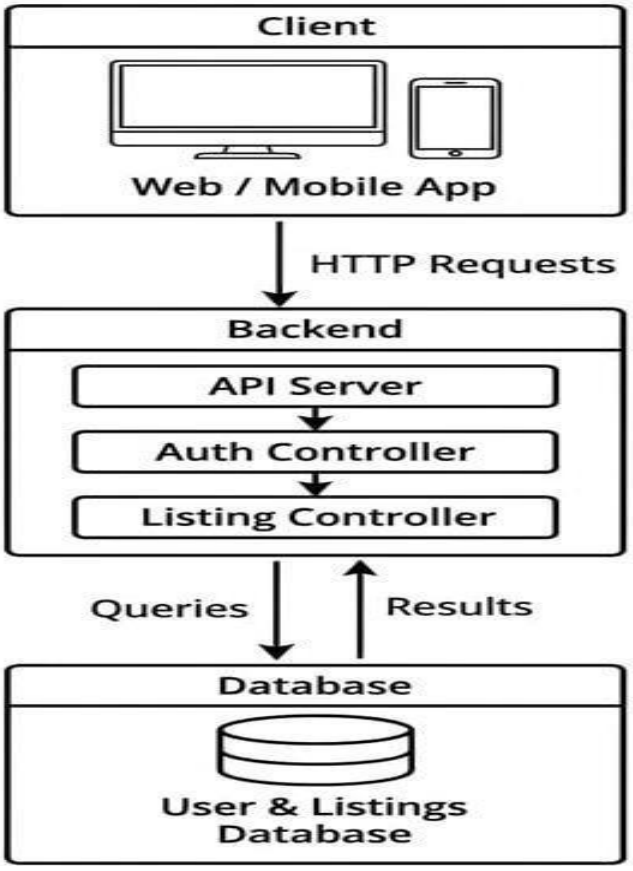

Fig 5.2 System Architecture Diagram

The System Architecture Diagram represents the overall structure of the Roommate Finder Website. It shows how different users interact with the system and how the system is organized in a structured manner. The architecture consists of three main layers: Client Layer, Backend Layer, and Database Layer.

The system starts with the client side where users access the application using a web browser or mobile device. The user sends HTTP requests to the backend server. The backend processes the request using API controllers such as authentication controller and listing controller. After processing, the server communicates with the database and returns the results to the user.

The architecture ensures proper separation of functionalities, secure data handling, and smooth interaction between users and the system.

Modules in System Architecture

1. **User Module**

- Register and login into the system
- Create and update profile
- Search available rooms
- View room details
- Contact room owner or Roommate

2. **Room Owner Module**

- Create profile and preferences
- Post room listings
- Manage room details
- View interested users

3. **Roommate Module**

- Search available listings
- View Roommate preferences
- Contact room owner
- Update personal preferences
- View room details

4. **Admin Module**

- Manage users and profiles
- Verify room listings
- Remove fake or invalid posts
- Manage database records

- **Key Features of Architecture**

- Role-based access control system
- Centralized database management
- Secure and structured data flow
- Easy navigation through modules
- Efficient communication between users

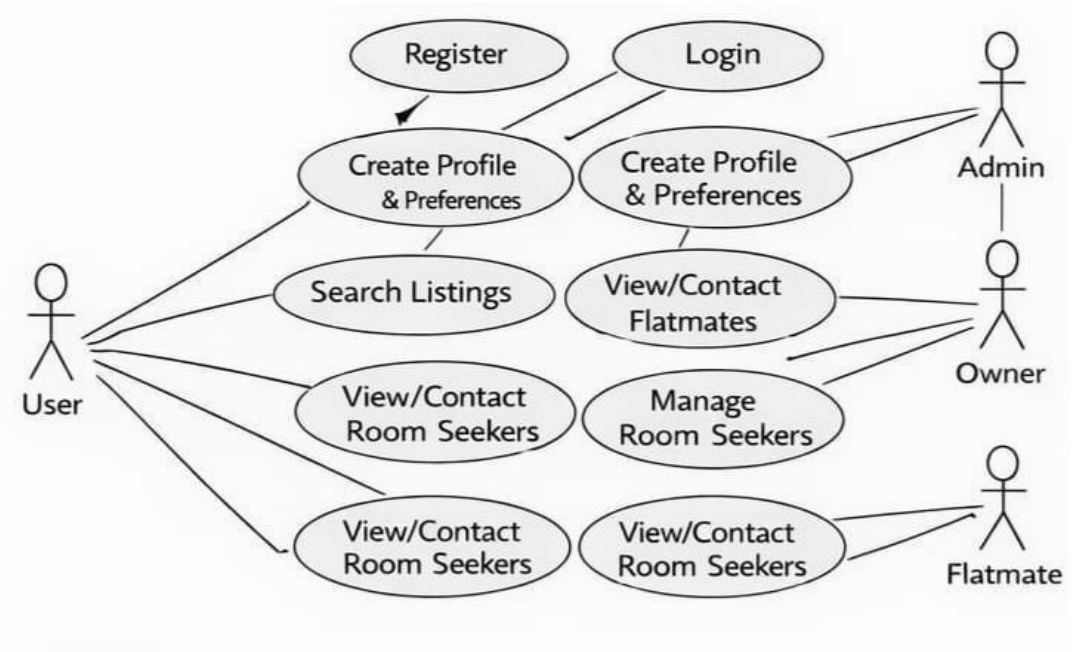

Fig. 5.3 Use Case Diagram

- **Actors**
  - User: Represents person searching for room or Roommate
  - Owner: User who posts room availability
  - Roommate: User searching shared accommodation
  - Admin: Manages entire system

- **Use Cases**

1. Authentication

• Register: User creates account in system

• Login: User logs into system

• Profile Creation: User adds personal details

• Verification: System validates user credentials

2. User Activities

• Search Listings: User searches available rooms

• View Room Details: User checks room information

• Contact Owner: User contacts room owner

• Create Preferences: User sets room preferences

• View Roommate: User views available Roommate

3. Owner Activities

• Create Listing: Owner posts room details

• Manage Listings: Owner updates room information

• View Room Seekers: Owner sees interested users

• Contact Users: Owner communicates with seekers

4. Roommate Activities

• Search Rooms: Roommate searches shared rooms

• View Preferences: Roommate checks compatibility

• Contact Owner: Roommate contacts room owner

• Update Profile: Roommate updates personal details

5. **Admin Activities**

- Manage Users: Admin controls user accounts
- Verify Listings: Admin approves room posts
- Remove Listings: Admin deletes invalid posts
- Monitor System: Admin monitors activities
- Generate Reports: Admin creates reports

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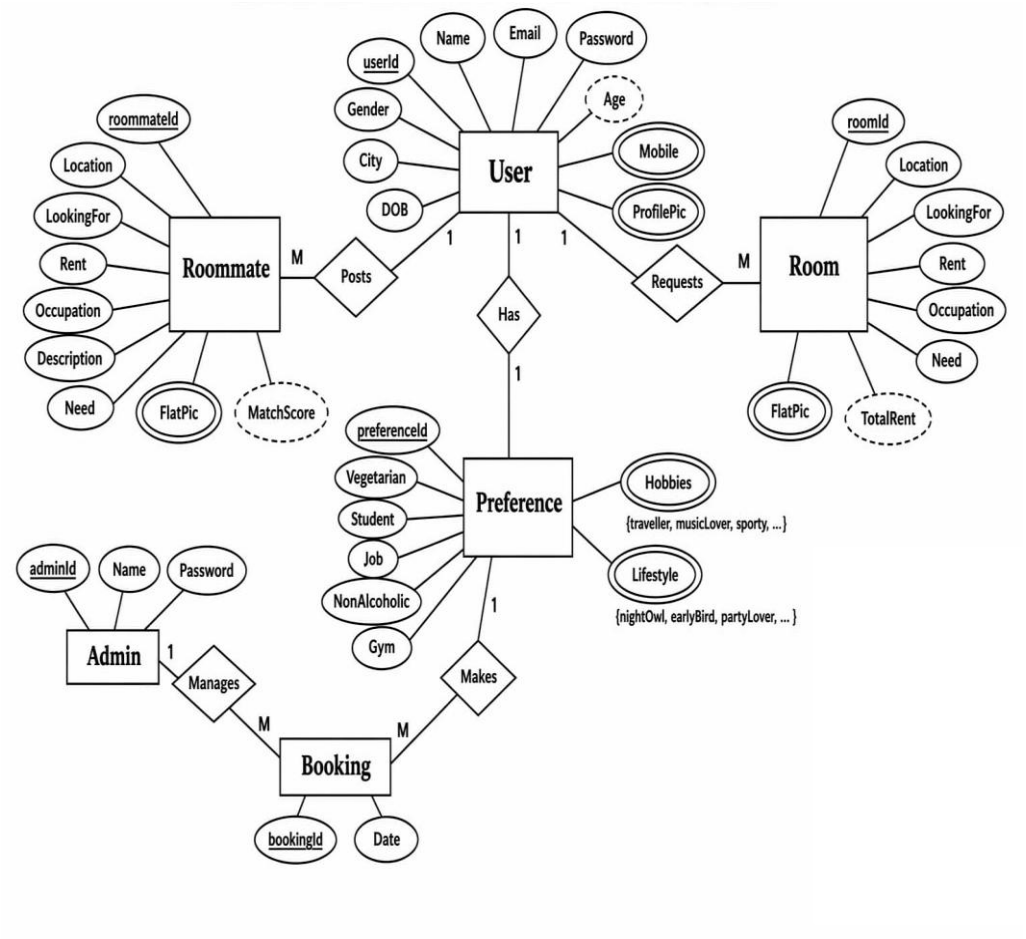

Fig. 5.4 Roommate Finder Website ER Diagram

The above ER diagram represents the Roommate Finder Website**,** which includes four main entities: User, Roommate, Room, and Preference The user entity stores details such as user id, name, gender, city, mobile number, and profile picture. The roommate entity contains information about users who are looking for roommates, including location, rent, occupation, description, and flat picture, while the Room entity stores room details like room id, location, rent, occupation, and description. The Preference entity includes user lifestyle choices such as vegetarian, student, job, gym, night owl, early bird, traveller, party lover, pet lover, and music lover. The relationships post request**,** and has define how the user can post roommate requirements, request rooms, and manage personal preferences within the system.

Chapter 6

Results and Applications

The Roommate Finder Website has been successfully developed and implemented as a web- based application to help users find rooms and Roommate efficiently. The system provides a digital platform for users, room owners, and administrators to interact and manage room listings in a structured and organized manner.

The application is developed using HTML, CSS, JavaScript, React and Node.js, with a backend database managed using Mongo DB. The system ensures easy accessibility, secure data handling, and efficient management of room and Roommate information.

1. **User Module Results**

The user module allows users to search and find rooms easily.

- Users can register and login into the system
- Users can search available rooms
- View room details such as rent and location
- Contact room owner or Roommate
- Compare different room options

This module helps users find suitable accommodation quickly and reduces manual searching effort.

2. **Room Owner Module Results**

The room owner module helps users post room availability.

- Owners can register and login
- Post room details with rent and facilities
- Update or delete room listings
- View interested users
- Contact room seekers

This module helps owners find tenants or roommate easily.

3. Roommate Module Results

The Roommate module helps users find shared accommodation.

- Users can create profile and preferences
- Search shared rooms
- Contact room owners
- Update preferences anytime

This improves communication and helps users find compatible roommate.

4. Admin Module Results

The admin module provides complete control over the system.

- Admin manages users and profiles
- Admin verifies room listings
- Admin removes fake or invalid posts
- Admin monitors system activities
- Admin manages database records

Admin ensures proper functioning, reliability, and security of the system.

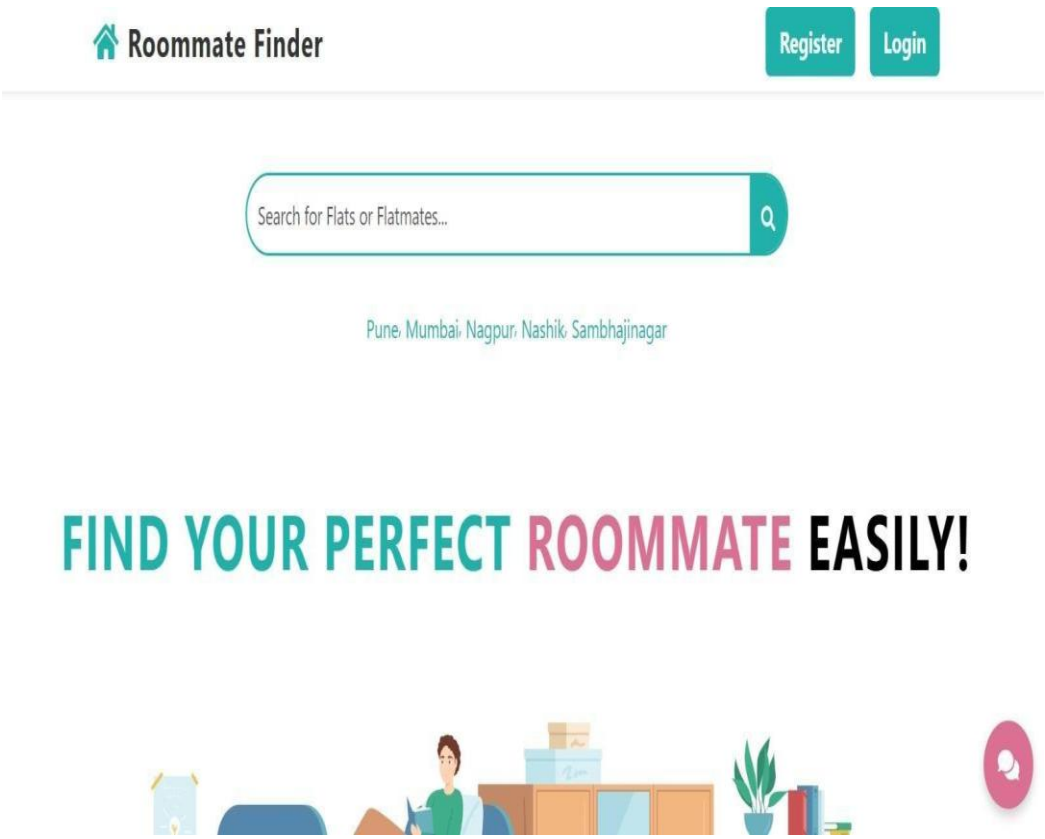

Fig 6.1: Home Page

The above diagram represents the Home Dashboard of the Roommate Finder Website. It provides a user-friendly interface for users to search for rooms or roommate easily. The dashboard displays important options such as search functionality, registration, and login for accessing system features.

At the top of the dashboard, a navigation bar is available which includes the project title and options like Register and Login. These options help new users create an account and existing users access their profiles. The simple navigation improves usability and allows users to move easily within the system.

The main section of the dashboard contains a search bar where users can search for rooms or roommate. Users can enter keywords such as location or preferences to find suitable accommodation. Below the search bar, popular cities are displayed, which helps users quickly choose their preferred location.

Fig 6.2 Popular Cities Rooms Listing page.

The above diagram represents the View Rooms in Popular Cities interface of the Roommate Finder Website. This interface allows users to easily explore available rooms and flats in different cities such as Pune, Mumbai, Nagpur, Nashik, Sambhajinagar, and Dhule. It provides a visual layout of city options, enabling users to quickly select their preferred location.

The interface is designed to be simple and user-friendly, where each city is represented with an image and name for better understanding. By clicking on a specific city, users can view detailed room listings available in that area. This feature helps users to search rooms efficiently based on location and improves the overall user experience in finding suitable accommodation.

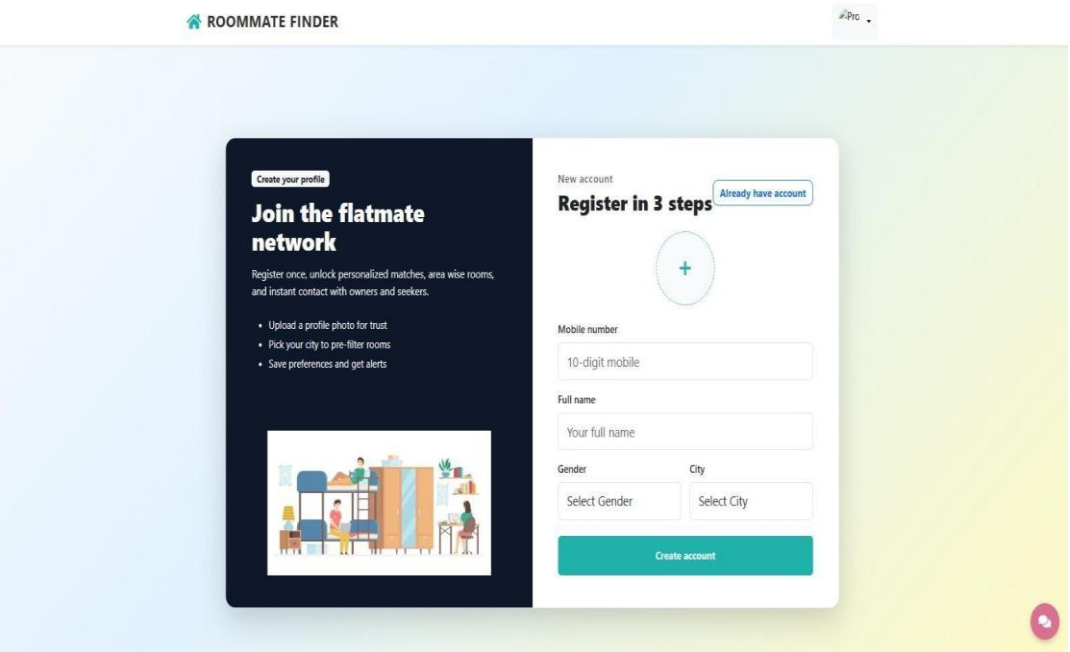

Fig 6.3: User register Page

The above diagram represents the User Registration interface of the Roommate Finder Website. This interface allows new users to create an account by entering their basic details such as mobile number, full name, gender, and city. It provides a simple form layout for easy data entry.

The interface is designed to be user-friendly and ensures that users can quickly register into the system. After filling in the required details, users can click on the register button to complete the registration process. This feature helps in managing user information and enables users to access various functionalities of the system such as searching and posting rooms..

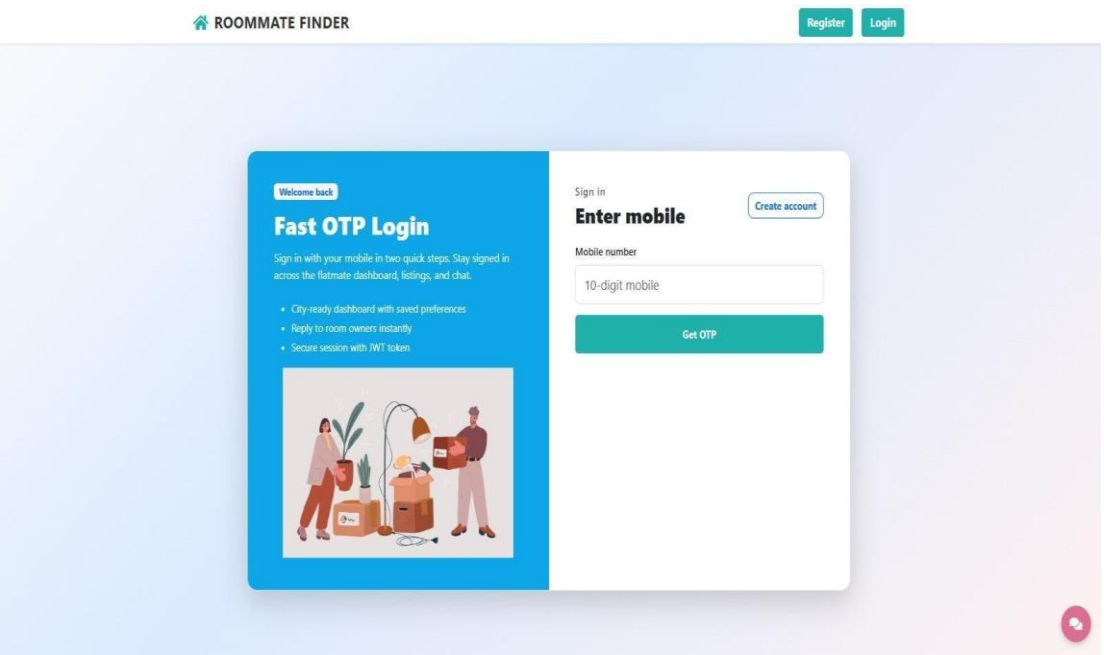

Fig 6.4: User Login Page

The above diagram represents the User Login interface of the Roommate Finder Website. This interface allows registered users to access the system by entering their mobile number. It uses an OTP (One-Time Password) based authentication method to ensure secure login.

The interface is simple and user-friendly, providing an input field for the mobile number and a button to generate OTP. After entering the correct OTP, users can successfully log in to the system. This feature ensures secure access and helps users to use services like searching rooms, posting rooms,

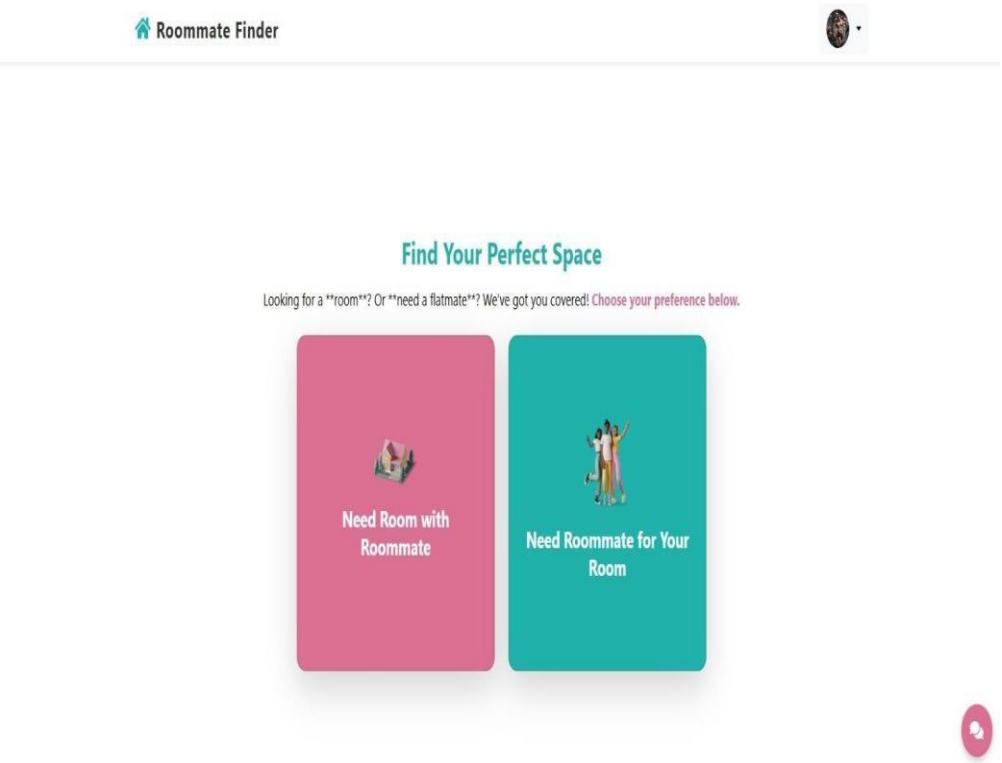

Fig 6.5: Choice Page

The above diagram represents the Home interface of the Roommate Finder Website. This interface provides users with two main options based on their needs: finding a room with a roommate or finding a roommate for their own room.

The interface is designed to be simple and visually clear, with two selectable sections that guide users to their desired action. Users can choose the appropriate option depending on whether they are searching for accommodation or offering a room. This feature improves user experience by providing a clear and

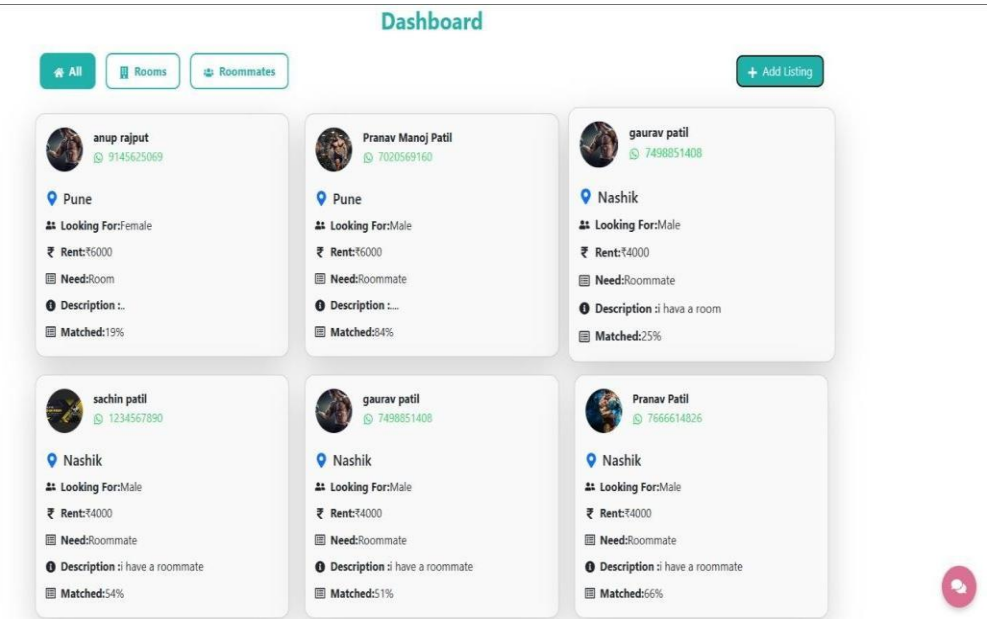

Fig 6.6: Dashboard Interface

The above diagram represents the Dashboard interface of the Roommate Finder Website. This interface displays multiple user listings in a card format, allowing users to easily browse available roommates and rooms.

Each card contains important details such as user name, contact information, location, preferred roommate type, rent amount, requirement (room or roommate), description, and matching percentage. This helps users quickly compare different options and find suitable matches.

At the top, the interface provides filtering options such as All, Rooms, and Roommates, which allow users to sort listings based on their preferences. There is also an “Add Listing” button that enables users to post their own room or roommate requirement.

The design is clean and user-friendly, making it easy to navigate through multiple listings. This interface improves user experience by presenting all necessary information in a structured and visually organized.

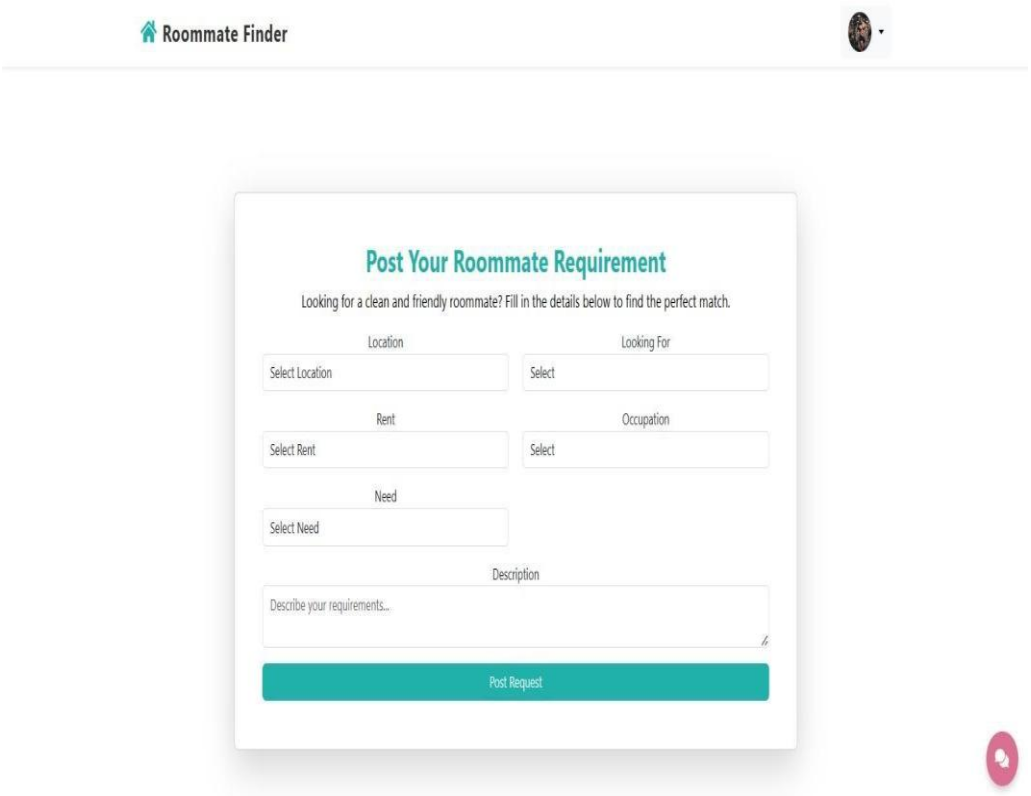

Fig 6.7: Post Roommate Requirement

The above diagram represents the Post Roommate Requirement interface of the Roommate Finder Website. This interface allows users to create and submit their roommate or room requirements by filling in necessary details.

The form includes input fields such as location, rent, need (room or roommate), preferred roommate type, and occupation, along with a description section where users can provide additional information about their preferences. These fields help in collecting accurate data for better matching.

Users can easily select options from dropdown menus, making the process simple and user-friendly. After filling in all the details, users can click on the “Post Request” button to submit their requirement.

The interface is designed in a clean and organized manner, ensuring that users can quickly enter their information without confusion. This feature improves the overall functionality of the system by enabling users to share their needs efficiently and get better roommate matches.

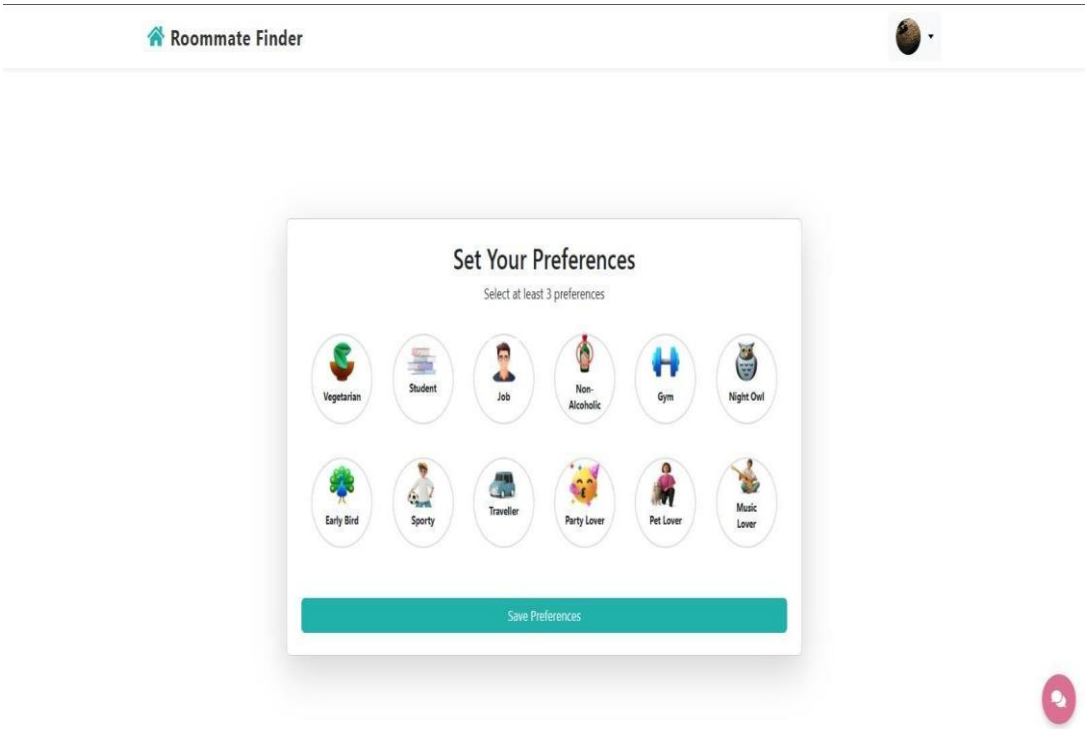

Fig 6.8: Set Your Preferences

The above diagram represents the Set Preferences interface of the Roommate Finder Website. This interface allows users to select their personal lifestyle and habit preferences to find a compatible roommate.

The screen displays multiple selectable options such as Vegetarian, Student, Job, Non-Alcoholic, Gym, Night Owl, Early Bird, Sporty, Traveller, Party Lover, Pet Lover, and Music Lover. Users are required to select at least a few preferences, which helps the system understand their personality and living style.

Each option is visually represented with icons, making the interface attractive and easy to use. Users can simply click on the options that match their preferences.

After selecting the desired preferences, users can click on the “Save Preferences” button to store their choices. These preferences are then used by the system to provide better and more accurate roommate matching.

The interface is clean, interactive, and user-friendly, enhancing the overall experience by allowing

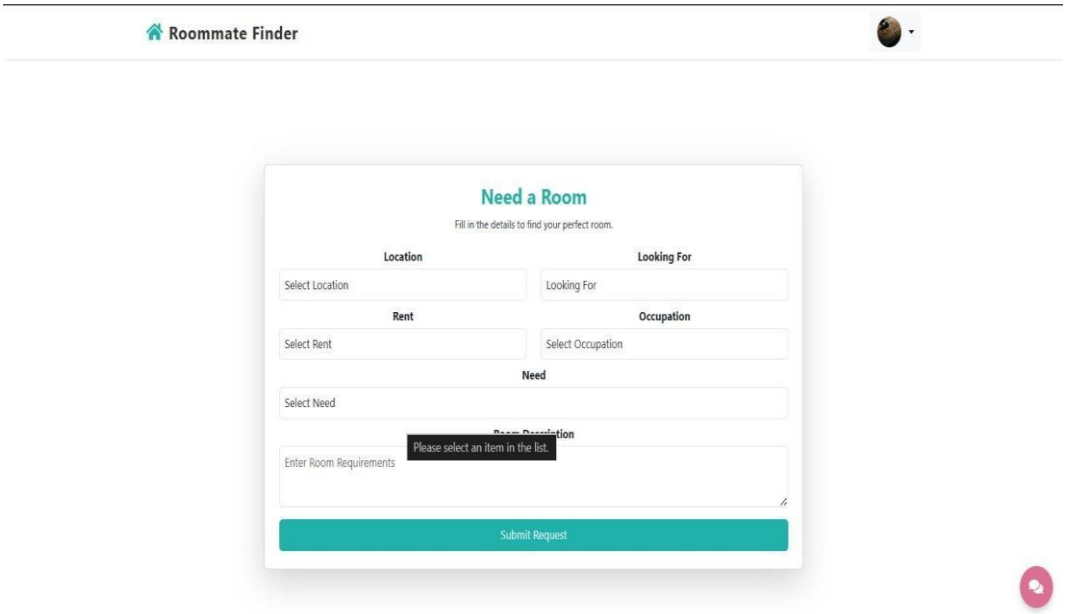

Fig 6.9: Need a Room Interface

The above diagram represents the Need a Room interface of the Roommate Finder Website. This interface is designed for users who are searching for a room based on their preferences.

The form includes fields such as location, rent, preferred roommate type, occupation, and specific needs, allowing users to provide detailed requirements. There is also a description section where users can enter additional information about the type of room or roommate they are looking for.

Users can select values from dropdown menus, making the process simple and structured. If any required field is not selected, the system displays a validation message (e.g., “Please select an item in the list”), ensuring proper data entry.

After filling in all the details, users can click on the “Submit Request” button to send their request. The system then uses this information to find and suggest suitable room options.

The interface is clean, organized, and user-friendly, helping users easily input their requirements and improving the overall efficiency of the room search process

Chapter 7

Testing and Defect Report

1. Introduction

The Test Plan for Roommate Finder Website outlines the overall strategy, scope, objectives, and methodology for testing the system. Testing is a crucial phase in the software development lifecycle, as it ensures that the system functions correctly, meets user requirements, and is free from critical defects. This test plan provides a structured approach to verify and validate the system functionalities.

2. Objectives of Testing

The main objectives of testing the system are as follows:

- To verify that all modules of the system are working as expected
- To identify defects, errors, and inconsistencies in the system
- To ensure that the system meets functional and user requirements
- To validate data accuracy and system performance
- To improve system reliability, usability, and efficiency

3. Scope of Testing

The testing process covers all major modules of the Roommate finder including:

- User Registration and Login Module
- Baby Profile Management Module
- Appointment Booking and Scheduling Module
- Vaccination Records Management Module
- Dashboard and Reporting Module

4. Testing Methodology / Approach

The following testing approach was used:

- Manual Testing: All functionalities were tested manually using predefined test cases
- Functional Testing: Each feature was tested to ensure correct operation
- Validation Testing: Input fields were tested with valid and invalid data
- Integration Testing: Interaction between modules was verified

5. Test Environment

The testing was performed in the following environment:

- Operating System: Windows 10/11
- Frontend Technologies: HTML, CSS, JavaScript, React
- Backend Technology: Node.js
- Database: Mongo DB

This environment ensured proper execution and validation of all system functionalities.

6. Test Data

Various types of test data were used to validate system behavior:

- Valid login credentials
- Invalid username and password combinations
- Sample baby details (name, DOB, parent details)
- Appointment data (date, time, notes)
- Vaccination records (vaccine name, date, status)
- Edge cases such as empty fields, incorrect inputs, and duplicate entries

7. Entry Criteria

Testing was initiated based on the following conditions:

- Development of all modules was completed
- Database was properly configured
- Required test cases were prepared
- System was deployed in a testing environment

8. Exit Criteria

Testing was considered complete when:

- All major test cases were executed
- Critical and high severity defects were resolved
- System performance was stable

9. Risks and Challenges

During testing, the following risks and challenges were identified:

- Possibility of data validation errors
- Performance issues due to large data
- UI inconsistencies on different devices
- Dependency on internet and server availability

10. Limitations of Testing

- Testing was conducted in a controlled environment
- Real-time user load testing was limited
- Some minor UI issues may still exist
- External integrations were not fully tested

11. Conclusion

The Test Plan provided a systematic and structured approach for testing the Roommate Finder wbsite It helped in identifying defects, improving system performance, and ensuring that all functionalities work as expected. After successful testing and fixing of identified issues, the system was found to be reliable, efficient, and ready for deployment.

Testing and Validation

Testing is a crucial phase in the development of the Roommate Finder Website It ensures that the system functions correctly, meets user requirements, and is free from errors. Various types of testing were conducted to verify the performance, functionality, and reliability of the system. The main objective of testing is to identify defects and ensure smooth operation of the application.

- **Types of Testing Performed**

1) Unit Testing – Individual modules like login, registration, and appointment booking were tested separately
2) Integration Testing – Interaction between modules such as patient, doctor, and staff was tested
3) System Testing – Complete system functionality was tested User Acceptance Testing – System was tested by users for real-time validation

Table No: 7.1 Test Cases

Test Case ID	Module	Description	Input	Expected Output	Actual Output	Status
TC01	Login	Valid login credentials	username & password	Login successful	Login successful	Pass
TC02	Login	Invalid password	Wrong password	Error message	Error displayed	Pass
TC03	Registration	Empty fields	blank input	Validation error	Error shown	Pass
TC04	User Module	Add user details	valid data	Data saved	Data saved	Pass
TC05	User Module	Missing email	empty email	Error message	Error shown	Pass
TC06	Room Post	Add room details	valid data	Room added	Added	Pass
TC07	Room Post	Missing rent	empty rent	Not allowed	Error shown	Pass
TC08	Room Search	Search rooms	valid location	Room list displayed	Displayed	Pass
TC09	Room Post	Missing location	empty field	Error	Accepted	Fail
TC10	Dashboard	Load dashboard	user login	Data displayed	Loaded	Pass
TC11	Contact	Contact owner	valid message	Message sent	Sent	Pass
TC12	Admin	Verify listing	valid listing	Approved	Approved	Pass
TC13	Profile	Update profile	valid input	Updated	Updated	Pass
TC14	Login	SQL injection test	malicio us input	Blocked	Blocked	Pass
TC15	Room Post	Duplicate listing	same data	Not allowed	Allowed	Fail

Failed Test Cases and Defects

During the testing phase, several defects were identified in the system which affected functionality and user experience. These issues were analyzed carefully to improve system performance and reliability.

The room posting module allowed submission of listings with empty or incomplete fields in certain scenarios, indicating insufficient input validation. This could lead to incorrect or incomplete room information being stored in the system.

The system also permitted duplicate room listings for the same details, which could create confusion for users searching for rooms. This indicated the need for proper duplicate validation.

There was a minor delay observed while loading the dashboard, especially when handling larger data sets, affecting system responsiveness. This required optimization of database queries and UI rendering.

These defects were further reviewed and necessary corrective measures were taken to enhance the overall system performance and user experience.

Table No: 7.2 Defect Report

Defect ID	Module	Description	Severity	Status
D01	Room Post	System allowed submission of room listings with empty or missing fields due to improper validation	Medium	Fixed
D02	Room Post	System allowed multiple bookings for the same date and time slot, causing scheduling conflicts	High	Fixed
D03	Dashboard	Dashboard experienced slight delay while loading data, affecting user experience	Low	Fixed
D04	UI	User interface alignment issues observed on different screen sizes, especially in mobile view	Low	Pending

Testing Summary

The Roommate Finder Website was thoroughly tested using multiple test cases that covered all major functionalities of the application, including login, registration, room posting, room searching, profile management, and admin verification. The primary objective of testing was to ensure that the system operates correctly, efficiently, and according to user requirements.

During the testing process, most of the test cases were executed successfully, indicating that the core functionalities of the system are working as expected. The system was able to handle user inputs properly, perform required operations, and display accurate results in different modules.

However, a few defects were identified during testing. These issues included validation errors, duplicate listings, minor performance delays, and user interface inconsistencies. Each defect was carefully analyzed based on its severity and impact on the system. Appropriate corrective measures were taken to resolve critical and medium-level issues, while minor issues were noted for further improvement.

After implementing the necessary fixes, the system showed significant improvement in terms of performance, reliability, and usability. The final testing results indicate that the system is stable, user- friendly, and capable of handling real-time operations effectively.

Overall, the testing phase played a crucial role in enhancing the quality of the system and ensuring that it meets the desired functional and performance standards.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

In conclusion, the Roommate Finder Website provides an efficient and user-friendly platform for finding rooms and compatible Roommate. The system allows users to search rooms, post room availability, and contact room owners in an organized manner. It simplifies the traditional method of searching accommodation and reduces manual effort.

The system provides a centralized platform where all room listings are stored and managed properly. Users can easily view room details such as location, rent, and facilities before making a decision. The admin module ensures that only valid and verified listings are displayed, which improves reliability and trust among users.

This system reduces dependency on brokers and saves time for users searching for accommodation. It also helps room owners find suitable tenants or Roommate quickly. The simple interface and structured database management improve overall system efficiency. The application ensures secure data handling and smooth communication between users.

Overall, the Roommate Finder Website improves accommodation management, enhances user experience, and provides a reliable digital solution for finding rooms and Roommate efficiently.

8.2 Future Scope

Mobile Application Development

A dedicated mobile application for Android and iOS can be developed to make the system more accessible. Users will be able to search rooms, post listings, and contact Roommate directly from their smartphones. This will improve usability and allow users to receive instant updates anytime and anywhere.

Multi-Language Support

The system can be enhanced by adding multi-language support so that users

from different regions can use the platform easily. Languages such as English, Hindi, and regional languages can be included. This will make the system more user-friendly and increase its reach to a wider.

Chat and Messaging System

A real-time chat feature can be integrated into the system to allow direct communication between room owners and seekers. Users will be able to discuss rent, facilities, and preferences without sharing personal contact details. This will improve communication and make the process faster
and safer.

Map and Location Integration

The system can include map integration to show the exact location of rooms. Users will be able to view nearby areas, transport facilities, and landmarks. This will help users make better decisions while selecting a room based on location.

Advanced Search and Filter Options

Advanced filtering options such as budget range, gender preference, room type, and amenities can be added. Users will be able to narrow down search results according to their requirements. This will reduce search time and improve user experience.

User Rating and Review System

A rating and review feature can be added where users can provide feedback about room owners or Roommates. This will help new users choose reliable listings and improve trust in the system. It will also encourage users to maintain genuine and accurate listings.

Cloud-Based Storage and Scalability

The system can be migrated to cloud-based storage for better performance and scalability. Cloud hosting will allow the system to handle a large number of users and listings efficiently. It will also improve data security, backup, and

Chapter 9

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